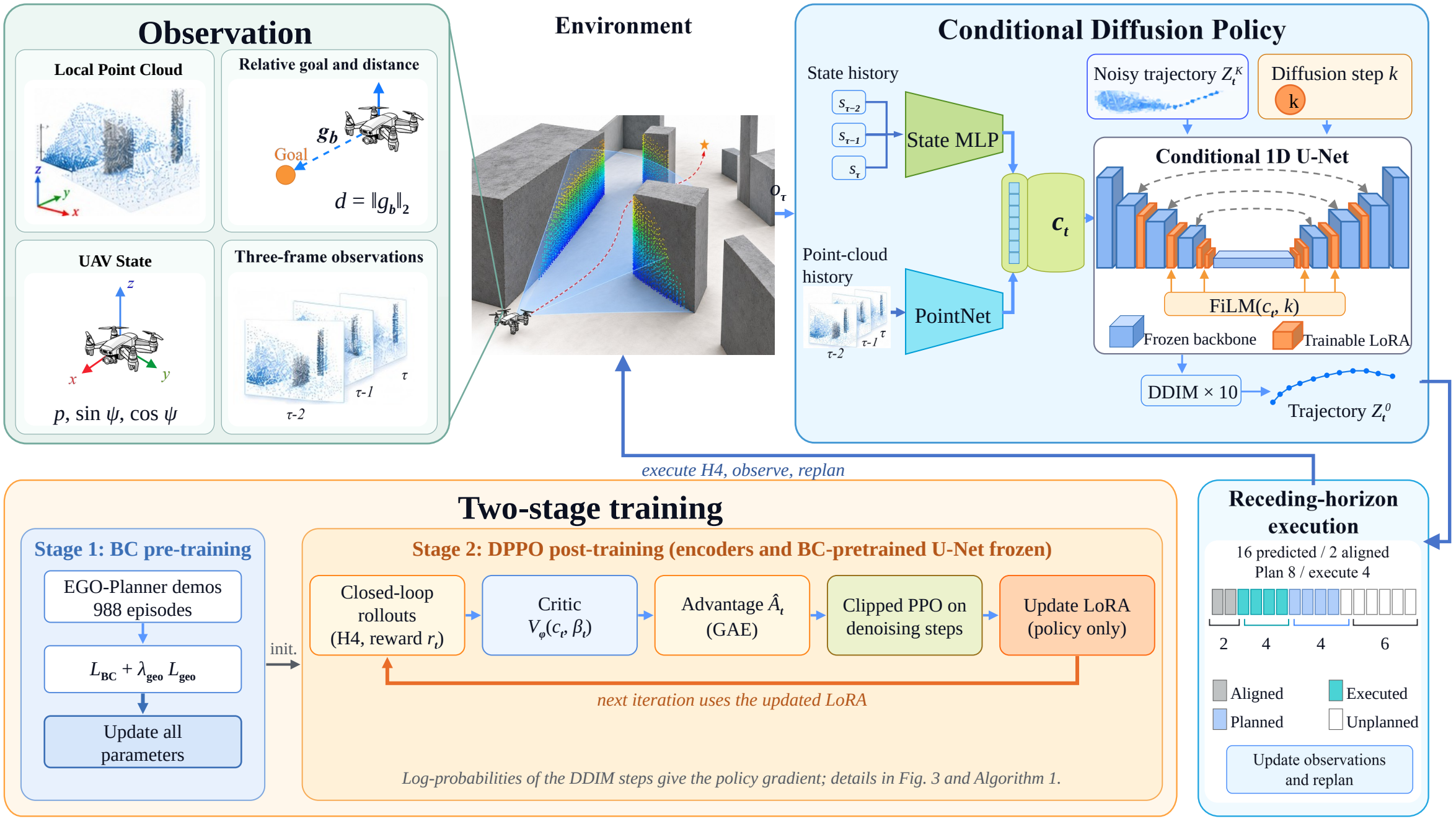
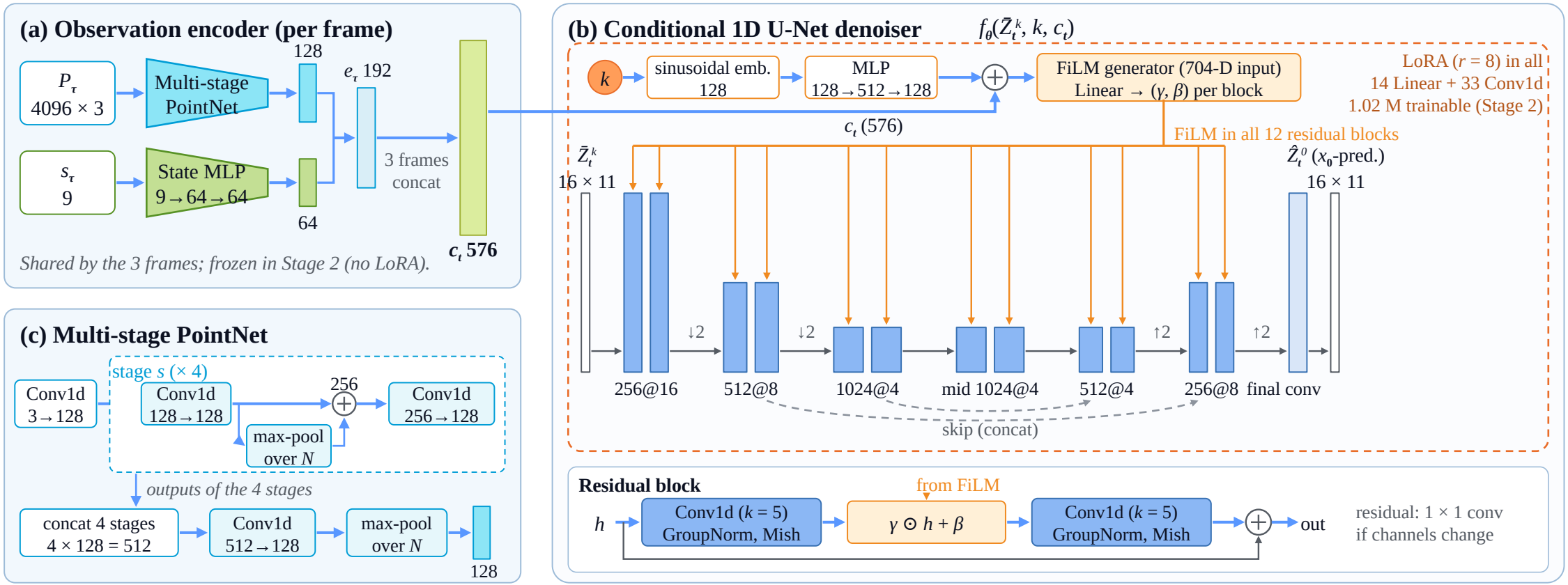
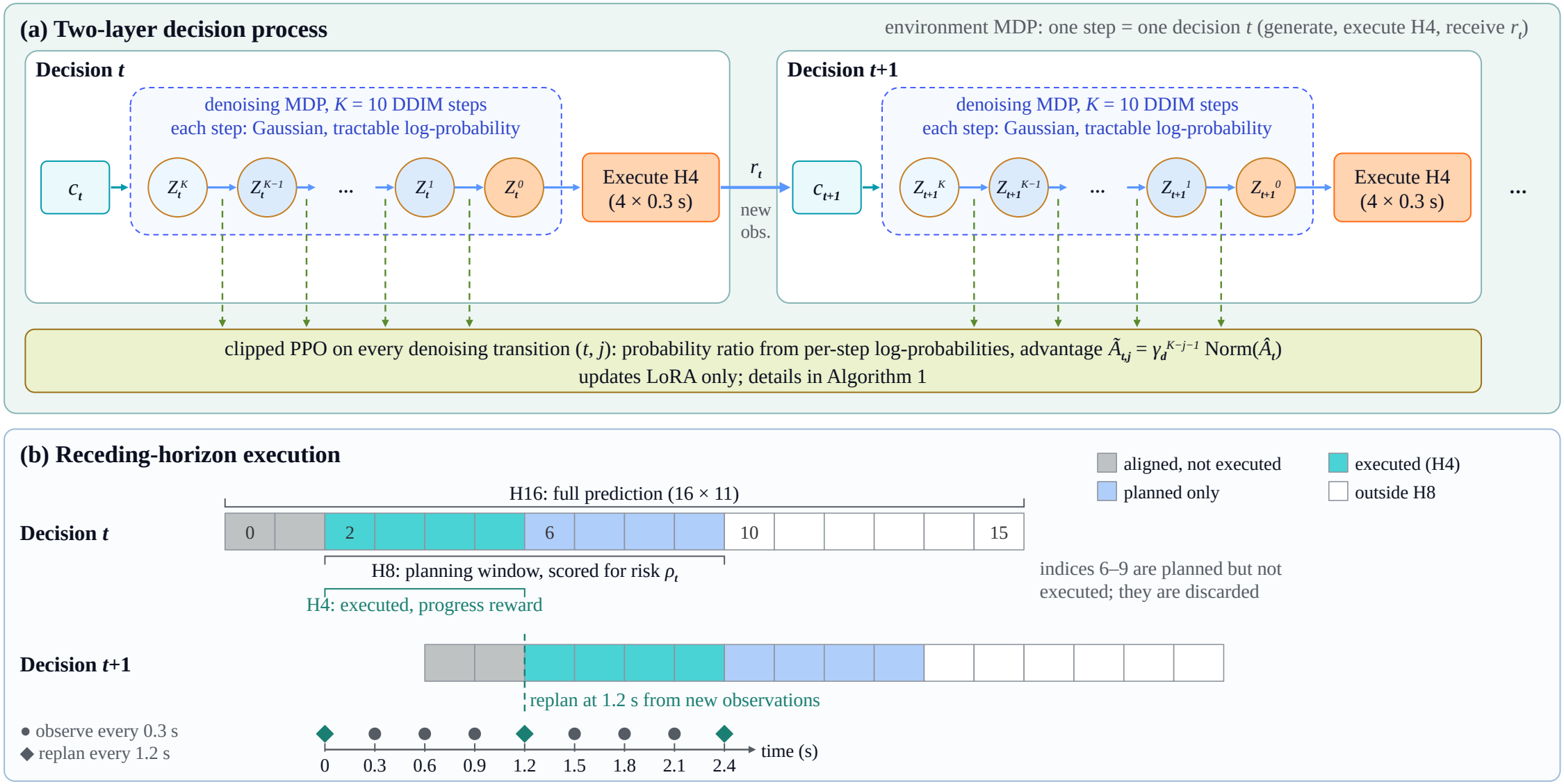

Observation
Local Point Cloud
Relative goal and distance
Goal
g_b
$d = \|g_b\|_2$
UAV State
p, sin ψ, cos ψ
Three-frame observations
τ-2
τ-1
τ
Environment
o_τ
Conditional Diffusion Policy
State history
$s_{\tau-2}$
$s_{\tau-1}$
s_τ
State MLP
Point-cloud history
PointNet
c_t
Noisy trajectory Z_t^K
Diffusion step k
k
Conditional 1D U-Net
FiLM(c_t, k)
Frozen backbone
Trainable LoRA
DDIM × 10
Trajectory Z_t^0
execute H4, observe, replan
Two-stage training
Stage 1: BC pre-training
EGO-Planner demos
988 episodes
$L_{BC} + \lambda_{geo} L_{geo}$
Update all parameters
init.
Stage 2: DPPO post-training (encoders and BC-pretrained U-Net frozen)
Closed-loop rollouts (H4, reward r_t)
Critic $V_\varphi(c_t, \beta_t)$
Advantage $\hat{A}_t$ (GAE)
Clipped PPO on denoising steps
Update LoRA (policy only)
next iteration uses the updated LoRA
Log-probabilities of the DDIM steps give the policy gradient; details in Fig. 3 and Algorithm 1.
Receding-horizon execution
16 predicted / 2 aligned
Plan 8 / execute 4
2
4
4
6
Aligned
Executed
Planned
Unplanned
Update observations and replan


(a) Observation encoder (per frame)
P_τ 4096 × 3
Multi-stage PointNet
128
e_τ 192
s_τ 9
State MLP 9 → 64 → 64
64
3 frames concat
c_t 576
Shared by the 3 frames; frozen in Stage 2 (no LoRA).
(b) Conditional 1D U-Net denoiser
$f_\theta(\bar{Z}_t^k, k, c_t)$
k
sinusoidal emb. 128
MLP 128 → 512 → 128
FiLM generator (704-D input) Linear → (γ, β) per block
LoRA (r = 8) in all 14 Linear + 33 Conv1d 1.02 M trainable (Stage 2)
c_t (576)
FiLM in all 12 residual blocks
$\bar{Z}_t^k$ 16 × 11
$\hat{Z}_t^0$ (x_0-pred.) 16 × 11
↓2
↓2
↑2
↑2
256@16
512@8
1024@4
mid 1024@4
512@4
256@8
final conv
skip (concat)
Residual block
from FiLM
h
Conv1d (k = 5) GroupNorm, Mish
$\gamma \odot h + \beta$
Conv1d (k = 5) GroupNorm, Mish
out
residual: 1 × 1 conv if channels change
(c) Multi-stage PointNet
Conv1d 3 → 128
stage s (× 4)
Conv1d 128 → 128
256
max-pool over N
Conv1d 256 → 128
outputs of the 4 stages
concat 4 stages 4 × 128 = 512
Conv1d 512 → 128
max-pool over N
128


(a) Two-layer decision process
environment MDP: one step = one decision t (generate, execute H4, receive r_t)
Decision t
denoising MDP, K = 10 DDIM steps
each step: Gaussian, tractable log-probability
c_t
Z_t^K
Z_t^{K-1}
...
Z_t^1
Z_t^0
Execute H4
(4 × 0.3 s)
r_t
new
obs.
Decision t+1
denoising MDP, K = 10 DDIM steps
each step: Gaussian, tractable log-probability
c_{t+1}
Z_{t+1}^K
Z_{t+1}^{K-1}
...
Z_{t+1}^1
Z_{t+1}^0
Execute H4
(4 × 0.3 s)
...
clipped PPO on every denoising transition (t, j): probability ratio from per-step log-probabilities, advantage $\tilde{A}_{t,j} = \gamma_d^{K-j-1}$ Norm($\hat{A}_t$)
updates LoRA only; details in Algorithm 1
(b) Receding-horizon execution
aligned, not executed
executed (H4)
planned only
outside H8
H16: full prediction (16 × 11)
Decision t
0
2
6
10
15
H8: planning window, scored for risk ρ_t
H4: executed, progress reward
indices 6–9 are planned but not
executed; they are discarded
Decision t+1
replan at 1.2 s from new observations
● observe every 0.3 s
◆ replan every 1.2 s
0
0.3
0.6
0.9
1.2
1.5
1.8
2.1
2.4
time (s)

Item	Learned policies (BC, post-trained)	EGO-Planner (controlled adaptation)
Tasks and initial states	84 original + 24 new tasks; frozen initial observations	Identical
Perception input	Synthetic depth camera; 4096 points per frame	Identical 4096-point observation
Goal	Read from the task list	Read from the task list (planner default replaced)
Global map	None; local observations only	No global map; map bounds enlarged to the task space, resolution unchanged
Clock	Simulation time, 0.3 s segments	Planner timers on simulation time, lock-step with the environment
Replanning	Every 1.2 s, after executing H4	Original timer and distance triggers
Planner parameters	—	Unchanged from demonstration collection
Termination and time limit	Collision, ground contact, arrival; 72 s	Identical evaluation code

identical condition controlled adaptation inherent difference, not aligned